

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Electronic Mail (E-mail)

Definition

Electronic Mail (E-mail) is an application-layer service that allows users to exchange messages over the Internet.

Characteristics

- One-way communication (sender may or may not receive a reply).
- Different from HTTP and FTP.
- Receiver does not need to remain online continuously.
- Uses intermediate mail servers for message delivery.
- Supports asynchronous communication.

Key Idea

Unlike HTTP and FTP, e-mail is not a direct client-server transaction between sender and receiver.

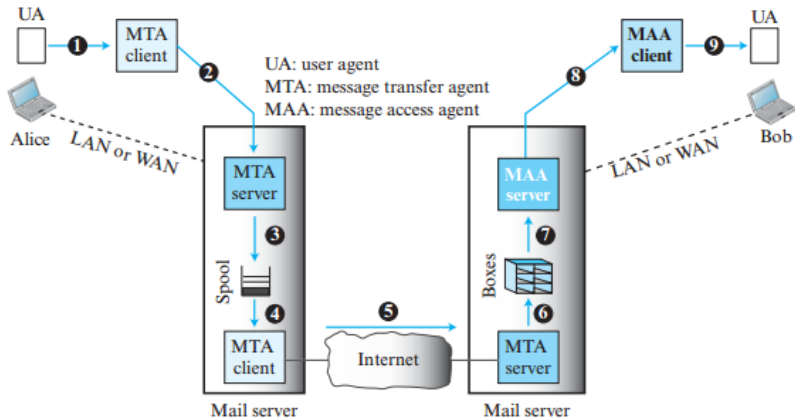
Components

- Sender (Alice)
- Receiver (Bob)
- Mail Server
- Mailbox
- Queue (Spool)
- Internet

Administrator creates:

- Mailbox for every user.
- Queue (Spool) for outgoing mails.

Figure 2.19 Common scenario



E-mail Architecture: Working

A simple e-mail from Alice to Bob involves **9 logical steps**.

Three Software Agents

- User Agent (UA)
- Message Transfer Agent (MTA)
- Message Access Agent (MAA)

Operation

- 1 Alice composes mail using UA.
- 2 Mail stored in sender's queue (spool).
- 3 MTA transfers mail through the Internet.
- 4 Bob's mail server stores message in mailbox.
- 5 Bob retrieves mail using MAA.

Exam Point

SMTP is a Push protocol.

MAA (POP3/IMAP4) is a Pull protocol.

User Agent (UA)

Definition

A User Agent (UA) is an application program that allows users to compose, send, receive, reply to, and forward e-mails.

Functions

- Compose e-mail
- Read received messages
- Reply and Forward
- Organize local mailbox
- Delete unwanted messages

Types of User Agents

- Command-driven User Agent
- GUI-based User Agent

Types of User Agents

Command-driven UA

- Keyboard-based
- Uses one-character commands
- Lightweight
- Used mainly in UNIX
- Examples:
 - mail
 - pine
 - elm

GUI-based UA

- Graphical Interface
- Easy to use
- Supports icons and menus
- Mouse and keyboard interaction
- Examples:
 - Outlook
 - Eudora

Sending an E-mail

A User Agent creates two components.

Envelope

Contains

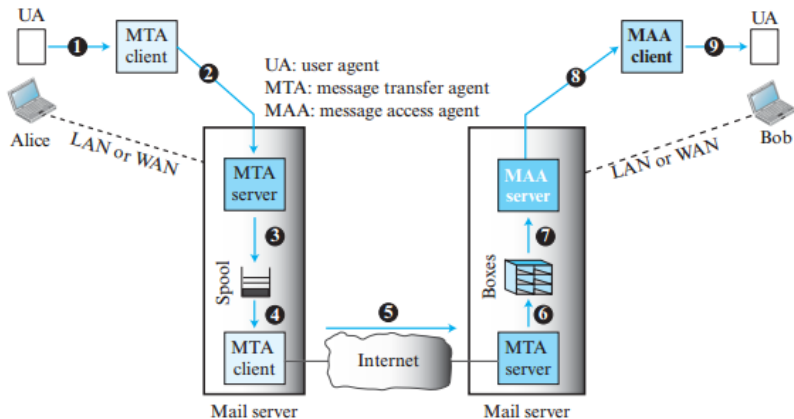
- Sender Address
- Receiver Address
- Routing Information

Message

Contains

- Header
- Body

Figure 2.19 Common scenario



Receiving E-mail

Receiving process

- 1 User Agent checks mailbox.
- 2 User receives notification.
- 3 Mail summary is displayed.
- 4 User selects a message.
- 5 Complete message is displayed.

Summary information includes

- Sender
- Subject
- Date and Time

E-mail Address Format

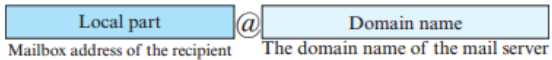
`local-part@domain-name`

- Local Part identifies the user's mailbox.
- Domain Name identifies the mail server.
- Every e-mail address is unique.

Mailing List (Alias)

- One alias represents multiple e-mail addresses.
- One message is automatically delivered to every member.
- Simplifies group communication.

Figure 2.21 *E-mail address*



Message Transfer Agent (MTA)

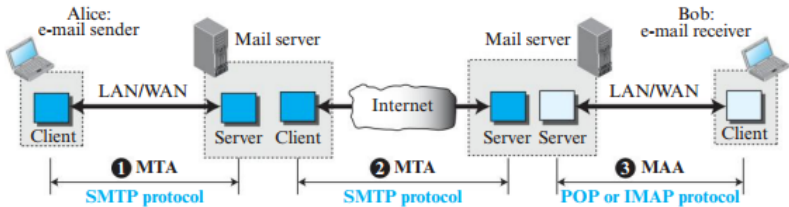
Definition

A Message Transfer Agent (MTA) transfers e-mails between mail servers using the **Simple Mail Transfer Protocol (SMTP)**.

Role of SMTP

- Transfers mail from Sender to Sender Mail Server.
- Transfers mail between Mail Servers.
- Uses the Client-Server model.
- Operates as a **Push Protocol**.
- Uses TCP Port 25.

Figure 2.22 *Protocols used in electronic mail*



SMTP Commands

SMTP communication takes place using commands sent by the client.

Command	Purpose
HELO	Identifies sender host
MAIL FROM	Specifies sender address
RCPT TO	Specifies recipient address
DATA	Starts message transfer
QUIT	Terminates SMTP session
RSET	Aborts current mail transaction
VERFY	Verifies recipient address
NOOP	Checks server status
EXPN	Expands mailing list
HELP	Displays help information

Exam Point

SMTP commands are sent from the client to the server.

SMTP Response Codes

Every SMTP command receives a response from the server.

Code	Meaning
220	Service Ready
221	Service Closing
250	Command Completed Successfully
251	User not local; message forwarded
354	Start Mail Input
421	Service Not Available
450	Mailbox Not Available
451	Local Error
452	Insufficient Storage
500	Syntax Error
501	Invalid Parameters
550	Mailbox Unavailable
551	User Not Local
552	Storage Exceeded
553	Mailbox Name Not Allowed
554	Transaction Failed

SMTP Mail Transfer Phases

SMTP transfers an e-mail in three phases.

① Connection Establishment

- TCP connection on Port 25
- Server sends 220
- Client sends HELO
- Server replies 250

② Message Transfer

③ Connection Termination

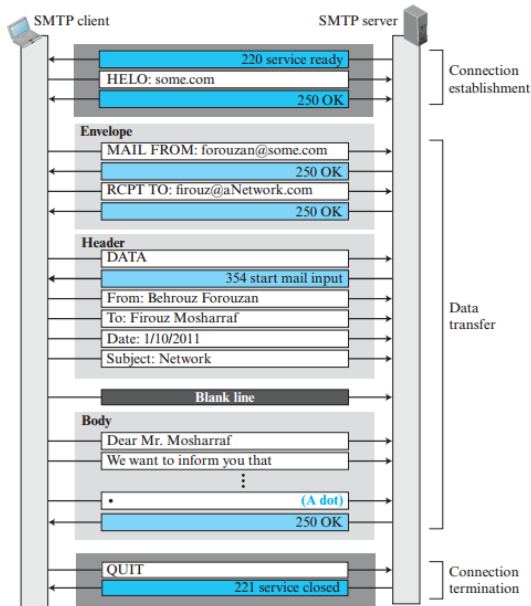
SMTP Message Transfer

Message Transfer Steps

- 1 MAIL FROM
- 2 Server replies 250
- 3 RCPT TO
- 4 Server replies 250
- 5 DATA
- 6 Server replies 354
- 7 Client sends Header and Body
- 8 Message ends with a single period (.)
- 9 Server replies 250
- 10 QUIT command closes the connection



Figure 2.23 Example 2.13



Message Access Agent (MAA)

Definition

SMTP is a **Push Protocol**. To retrieve e-mails from a mail server, a **Message Access Agent (MAA)** is required.

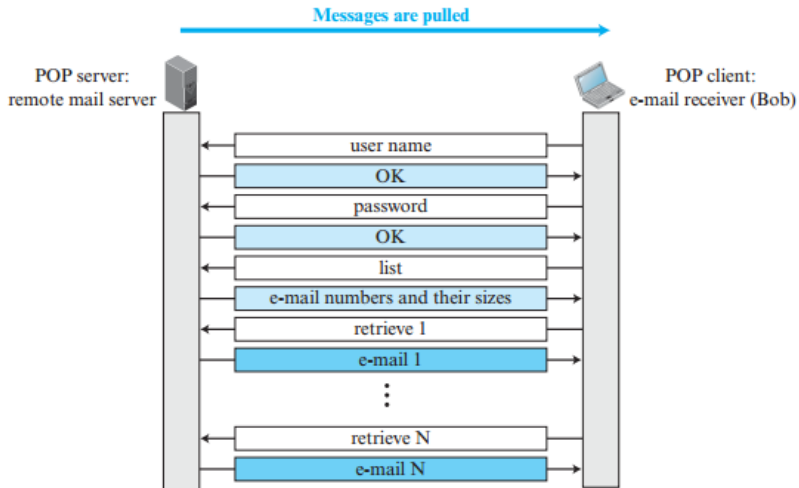
Popular MAA Protocols

- POP3 (Post Office Protocol Version 3)
- IMAP4 (Internet Mail Access Protocol Version 4)

POP3

- Uses TCP Port 110
- Downloads messages from the mail server
- Supports Delete Mode and Keep Mode

Figure 2.24 POP3



IMAP4 vs POP3

Feature	POP3	IMAP4
Mail Storage	Local Computer	Mail Server
Multiple Devices	Limited	Supported
Folders	No	Yes
Partial Download	No	Yes
Synchronization	No	Yes
Header Preview	No	Yes
Search Before Download	No	Yes

Advantage of IMAP4

Supports server-side folders, synchronization, partial downloads, header preview, and searching before downloading.

MIME (Multipurpose Internet Mail Extensions)

Need for MIME

SMTP supports only 7-bit ASCII text.

MIME enables transmission of multimedia and non-ASCII data.

Supports

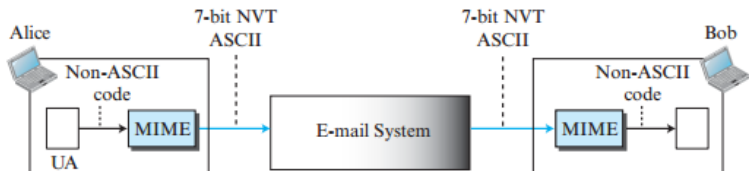
- Images
- Audio
- Video
- PDF and Binary Files
- Multiple Languages



Important MIME Headers

- MIME-Version
- Content-Type
- Content-Transfer-Encoding
- Content-ID
- Content-Description

Figure 2.25 *MIME*



Content Transfer Encoding

- 7-bit
- 8-bit
- Binary
- Base64
- Quoted-Printable



Web-Based Mail

- Gmail
- Yahoo Mail
- Outlook.com
- Uses HTTP/HTTPS between browser and mail server
- SMTP is still used between mail servers

Figure 2.28 *Quoted-printable*

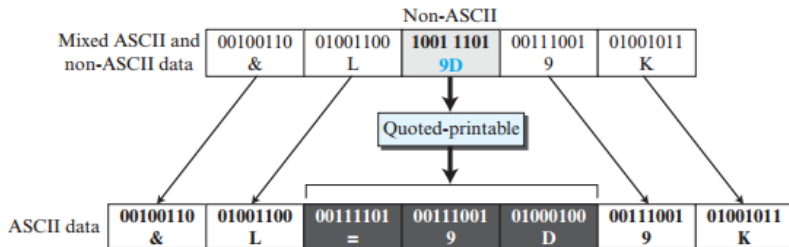
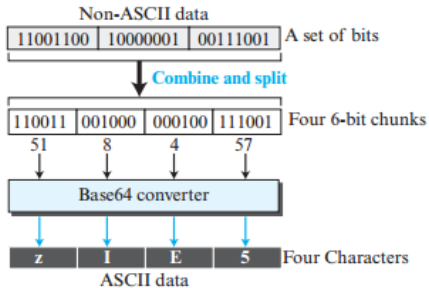


Figure 2.27 Base64 conversion



Security Issues

- SMTP does not provide built-in security.
- Messages may be intercepted.
- Passwords may be exposed.

Security Solutions

- Pretty Good Privacy (PGP)
- Secure/Multipurpose Internet Mail Extensions (S/MIME)
- Provide Confidentiality, Authentication, Integrity and Digital Signatures